

FILTER DESIGN TEST CASE

Lattice Filter - Band 66 (AWS) for IoT Narrowband Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting Band 66 (AWS) for IoT Narrowband Module. The filter is designed on PZT on Si substrate with a target center frequency of 5143.4 MHz and bandwidth of 279.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the IoT Narrowband Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: Band 66 (AWS)
- Application: IoT Narrowband Module
- Substrate: PZT on Si
- Package: LGA (Land Grid Array)
- Design Tool: Cadence AWR Microwave Office
- Design Center: Osaka Design Center

This specification applies to all variants of the Lattice Filter design targeting Band 66 (AWS). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5143.4 MHz
Bandwidth (3 dB):	279.7 MHz
Insertion Loss Target:	≤ 1.7 dB
Return Loss Target:	≥ 15.0 dB
Out-of-Band Rejection:	≥ 51.0 dB
Isolation (Duplexer):	≥ 27.0 dB
Q Factor:	3782
Temperature Coefficient:	-18.6 ppm/°C
Die Size:	1.2 x 0.9 mm
Wafer Size:	8-inch
Number of Resonators:	12
Electrode Material:	W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the Lattice Filter template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 5143.4 MHz, BW 279.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.7 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, PZT on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15430 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.7 dB
- Return loss in passband shall be >= 15.0 dB
- Out-of-band rejection at +/- 420 MHz offset >= 51.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 12.0 MHz
- Power handling shall be >= +33 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 85%

6. TEST RESULTS

Measurement Date: 2024-04-12
Devices Tested: 66
Insertion Loss (meas): 1.4 dB
Return Loss (meas): 17.4 dB
Rejection (meas): 53.9 dB
Isolation (meas): 32.3 dB
Q Factor (meas): 3782
Group Delay Variation: 3.8 ns
Power Handling: +31 dBm
Die Yield: 90.5%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Lattice Filter design demonstrated excellent performance for Band 66 (AWS).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Approver:	Dr. James Liu	Date:	2024-04-22

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